

PROJECT REPORT: Temporal Patterns in Urban Crime

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1. Executive Summary

This project investigates urban crime patterns using a dataset of over 1,000 incidents. The primary goal was to identify how time-based variables, such as the hour of the day and the day of the week—influence crime frequency and type.

Using Pandas for data manipulation and NumPy for statistical anomaly detection, the analysis revealed that crime is not evenly distributed across the clock.

Key findings indicate that the "Night" period accounts for the highest proportion of incidents and that specific high-density "Areas" exhibit crime volumes that are statistically significant outliers (Z-score > 1.5). These insights suggest that public safety resources should be prioritized based on temporal and geographic hotspots.

2. Introduction and Project Objectives

Understanding when and where crimes occur is essential for effective law enforcement and community safety planning. This project explores a real-world crime dataset to uncover hidden temporal trends.

Research Question:

Research Question 1: How does the frequency of crime vary across different times of day (Morning, Afternoon, Evening, Night)?

Research Question 2: Which geographic areas act as statistical "High Density Zones" based on Z-score analysis?

Research Question 3: Is there a visible relationship between the time of day and the specific types of crime committed?

3. Dataset Description

The dataset contains detailed logs of criminal incidents, including the date, time, location (Area), and victim demographics .

3.1 Basic Dataset Information

Property	Value
Dataset Source	City of Los Angeles - Crime Data from 2020 to 2024
Number of rows	134198
Number of columns	11
Data types present	Numerical, Categorical, and Date-Time
Target / Key variables	CRIME, TIME OF DAY, AREA NAME
Potential issues	Missing weapon data and non-standard gender entries

4. Data Cleaning and Preparation

Four major cleaning steps were performed to ensure the analysis was accurate .

Issue Found	Action Taken	Reason / Justification	Code Reference
Missing Occurrence Dates	Dropped rows with NaN in DATE OCC	Primary analysis is time-based; missing dates are unusable.	Cell 2
Inconsistent Gender Labels	Used .str.upper() and replaced "UNKNOWN"/"X" with np.nan	Standardizes categories for accurate demographic pivoting.	Cell 2

Issue Found	Action Taken	Reason / Justification	Code Reference
Missing Weapon Data	Filled NaN with "WEAPON UNKNOWN"	Prevents loss of data rows while acknowledging missing info.	Cell 2
Incorrect Date Types	Converted strings to <code>pd.to_datetime</code>	Enables extraction of month, year, and day names.	Cell 2

5. Feature Engineering

New features were derived to facilitate deeper analysis .

New Feature	How It Was Created	Why It Is Useful
HOUR OCC	Extracted the hour from the <code>TIME OCC</code> HHMM integer.	Allows for hourly trend analysis .
TIME OF DAY	A custom function grouping hours into Morning, Afternoon, Evening, and Night.	Simplifies complex hourly data into actionable shifts .
DAY OF WEEK	Extracted from <code>DATE OCC</code> using <code>.dt.day_name()</code> .	Enables comparison of weekday vs. weekend crime rates .

6. Analysis Methodology

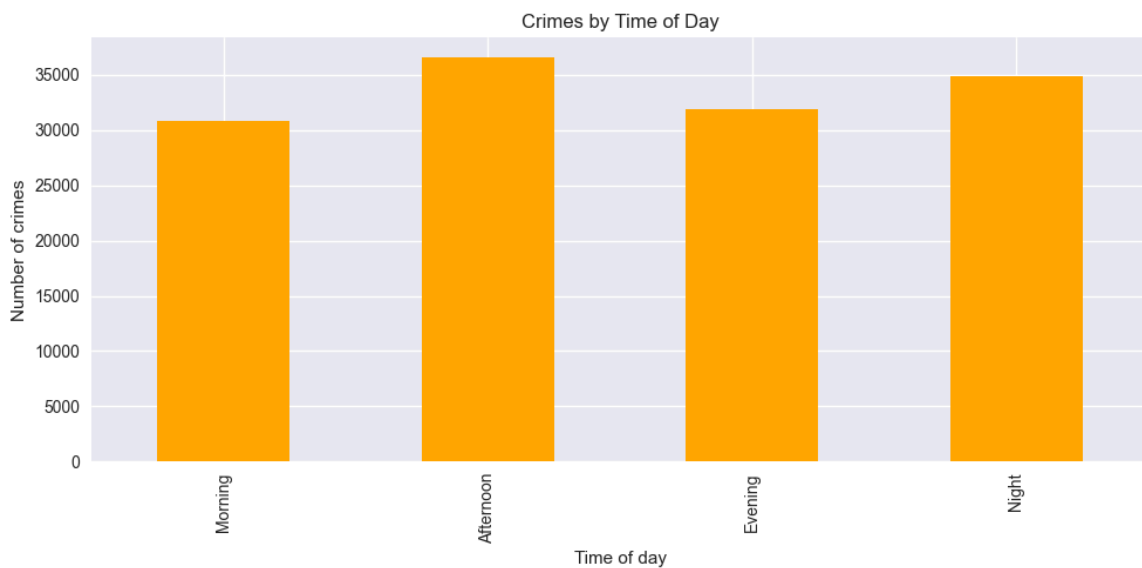
- **Pandas:** Used for loading the dataset, loading date formats, cleaning, grouping, and creating summary tables .
 - **NumPy:** Used for numerical computations including mean, standard deviation, and Z-scores to identify statistical anomalies .
 - **Matplotlib:** Used to create at least 4 meaningful visualizations supporting the research questions .
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7. Analysis and Visualizations

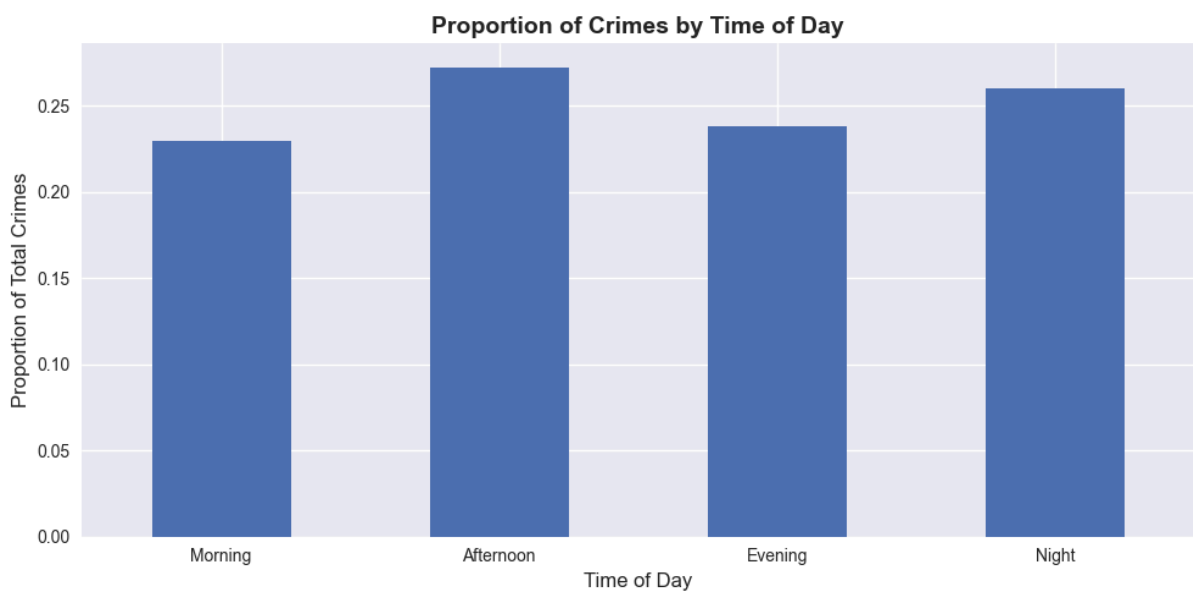
7.1 Research Question 1: Temporal Distribution

Analysis: We aggregated crime counts by the engineered TIME OF DAY feature .

- **Visualization:** Chart of crimes by the time of day.



Visualization 2: Time Proportion of crime

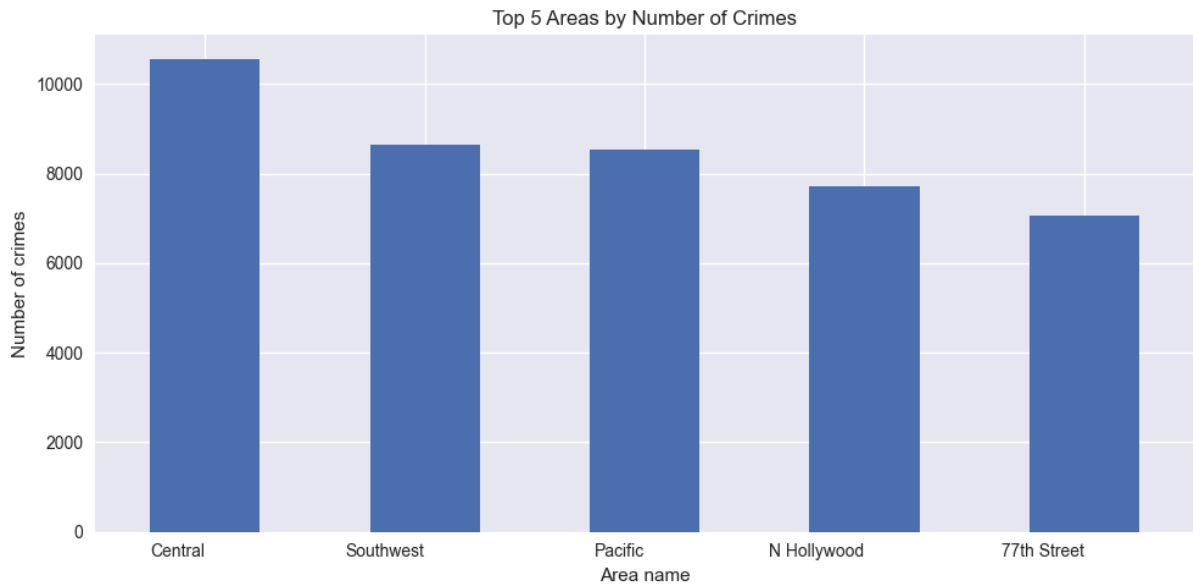


- **Interpretation:** The analysis indicates that the "Afternoon" and "Night" period is the most active for crime incidents .

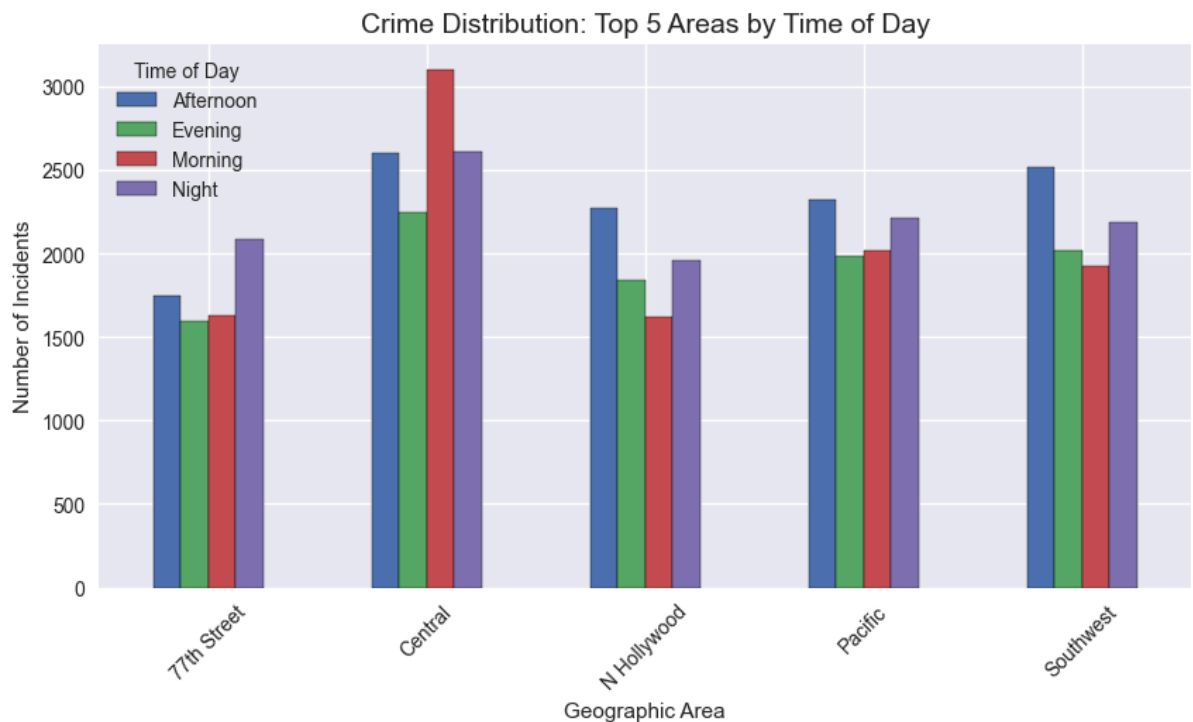
7.2 Research Question 2: Geographic Hotspots

Analysis: Applied Z-score analysis to identify high-density zones .

- **Visualization:** Top 5 Areas by the number of crimes



- **Visualization 2:** Top 5 Areas by hourly number of crimes.

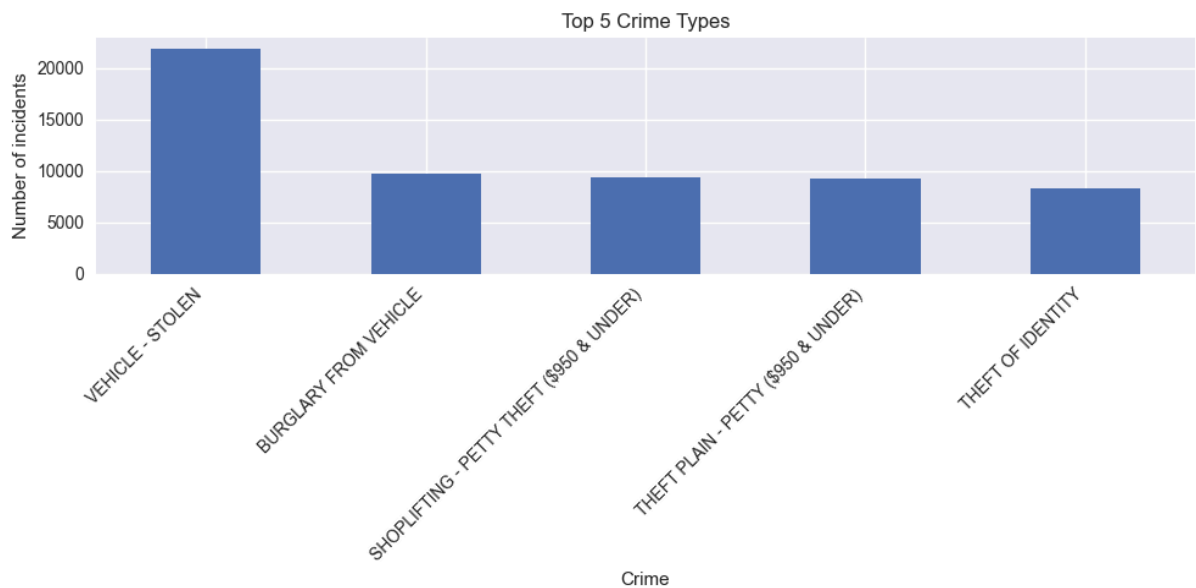


- **Interpretation:** Areas with a Z-score > 1.5 represent statistical high-crime density locations and the severity by the time of day.

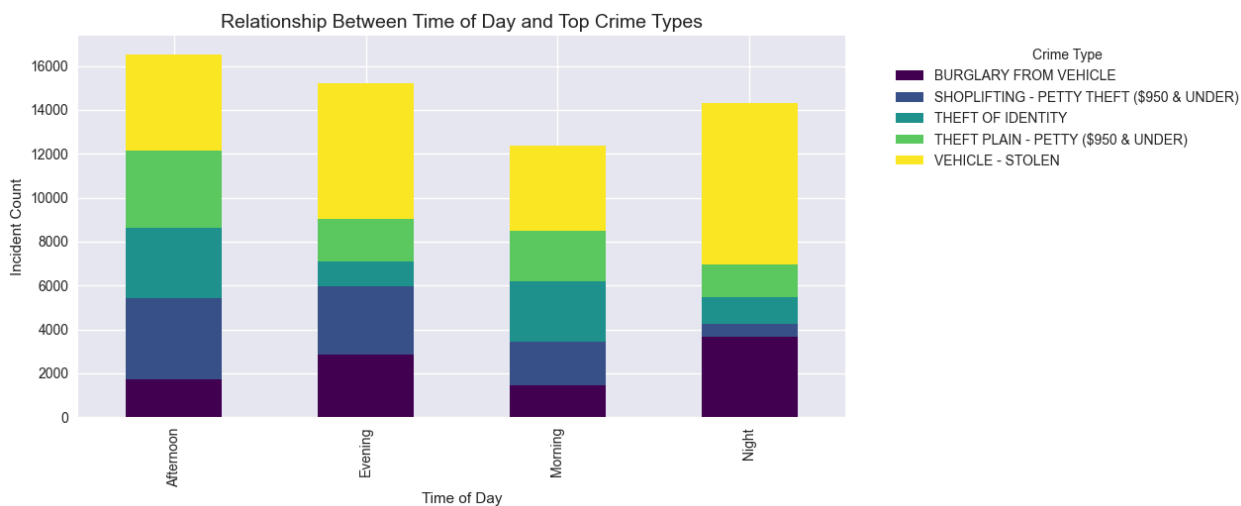
7.3 Research Question 3: Relationship Analysis

Analysis: Correlated Time of Day with Crime types using pivot tables and stacked bar charts.

● **Visualization 1:** bar chart for Top 5 Crime Types.



● **Visualization 2:** Stacked bar chart for Relationship Between Time of Day and Top Crime Types .



● **Interpretation:** Specific crime types vary significantly across different times of the day .

8. NumPy-Based Custom Computation

Component	Description
Computation used	Area Safety Z-Score Calculation
Purpose	To identify areas that are statistically anomalous compared to the mean .
Logic	$Z = (x - \mu) / \sigma$ using np.mean() and np.std() .
Result / Takeaway	Identified areas that act as high-density outliers .

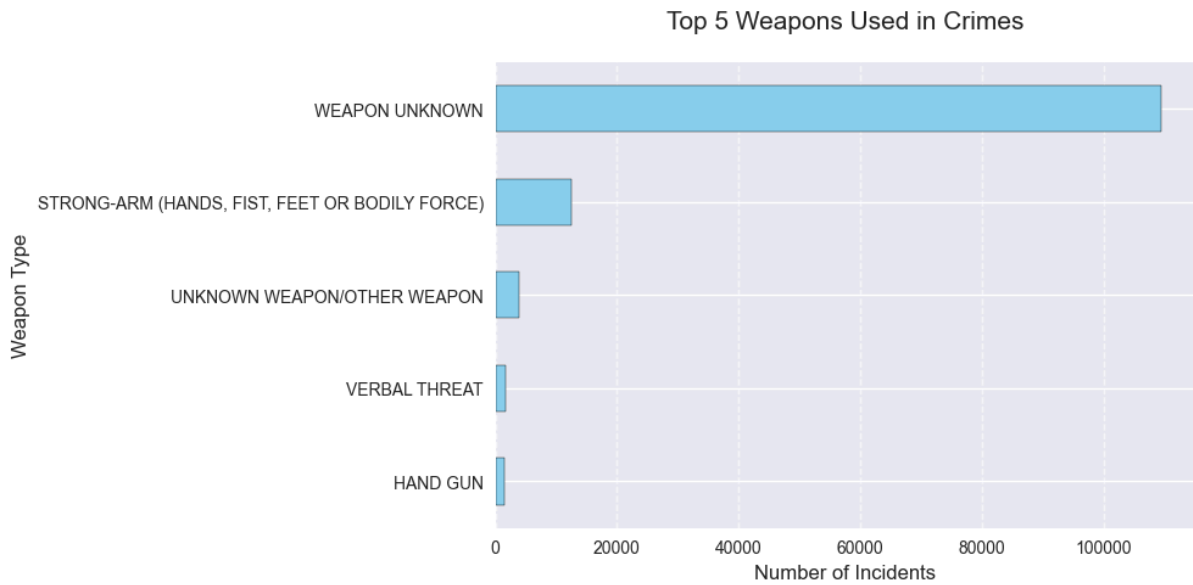
9. Key Findings

1. **Temporal Peak:** The "Night" period contains the highest total count of crimes .
 2. **High-Risk Areas:** Z-score analysis identifies specific areas as high-density zones .
 3. **Outlier Spikes:** Specific dates were identified as outliers with crime spikes beyond normal variations .
 4. **Weapon Trends:** Blunt objects and unknown weapons represent common choices in the top 5 weapon categories .
 5. **Gender Patterns:** Distinct crime patterns were visible based on victim sex through relationship analysis .
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10. Limitations

The project is subject to limitations such as reporting bias and incomplete data in columns like "WEAPON". Correlation between time and crime does not imply direct causation .

- **Visualization:** Top 5 Weapons used in crimes



11. Conclusion

The analysis successfully identifies key temporal and geographic patterns in urban crime . Future work could involve predictive modeling based on these identified hotspots .

12. References / Dataset Source

- Original Data Source. [City of Los Angeles - Crime Data from 2020 to 2024](#)
- Crime Data from 2024 to 2025. [python-project-2/Crime/Data/Crime_Data.csv at main · Ishtyahmd/python-project-2](#)